Persuasion for protection: an analysis of online safety videos on YouTube

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Liuyu Huang, Dion Hoe-Lian Goh and Stella Xin Yin
Wee Kim Wee School of Communication and Information, Nanyang Technological University, Singapore, Singapore

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Abstract

Purpose – Public service announcements (PSAs) have been shown to be effective instruments that raise awareness, educate society, and change behaviors and attitudes. Many governments and organizations have utilized PSAs on social media to promote online safety among children and youth. However, we have limited understanding of the range of topics that these PSAs address and how they present their content to audiences. This study provides an inventory of the types of online safety topics that current PSAs address and a catalogue of the types of persuasive features employed by PSAs.

Design/methodology/approach – A content analysis of 220 YouTube PSA videos on online safety was conducted. Various topics under the umbrella of online safety were identified. Guided by the prospect theory and exemplification theory, different persuasive features employed in the PSAs were sought.

Findings – The findings highlight that the primary focus of these PSAs is on online safety behaviors and general instructions on online hygiene. Interestingly, nearly half of the videos employ a neutral frame, while a significant portion provides no evidential support. Additionally, video length was associated with the number of views and likes it gathered but not with the number of comments.

Originality/value – The inventory of PSAs can help researchers, practitioners, and policymakers better understand the type of content being produced and disseminated online as well as identify topics that are either over or under-represented. Further, the catalogue of the types of persuasive features employed by PSAs would be helpful in guiding research, practice, and policymaking in the context of creating effective online safety videos.

Keywords Public service announcements, Online safety, YouTube videos, Persuasive features, Content analysis, Children and youth

Paper type Research paper

Introduction

Social media and Internet use are popular among children and youth. Their usage opens a wide range of possibilities for this age group, both for educational purposes and for leisure activities. However, this poses a problem: research shows that children and youth often do not have the capacity to engage with online services in a safe manner, especially when it comes to interactive social media (Li *et al.*, 2023; Reuters, 2023). For example, Lenzi *et al.* (2023) found that social media addiction does harm to students' school achievement, while Nazir *et al.* (2020) found that Malaysian children and youth spent too much time on social media resulting in their sleeping time being much less than recommended. Moreover, adolescents are characterized by risk-taking and independence-seeking from parents (Andrews *et al.*, 2021), and online interactions magnify their opportunities to engage in potentially detrimental activities (Rutkowski *et al.*, 2021). It is thus understandable that children and youth might suffer negative consequences when they are exposed to improper content on social media (Cataldo *et al.*, 2021).

In a parallel development, young people have increasingly used video-sharing platforms such as YouTube for information (Zimmermann *et al.*, 2020). This phenomenon naturally

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increases their exposure to possible improper content (Tahir *et al.*, 2019), which could result in physical or mental harm. For example, Khasawneh *et al.* (2020) found that most of the videos on the blue whale challenge, where young people were encouraged to engage in self-harm and eventually kill themselves, violated at least 50% of the Suicide Prevention Resource Center (SPRC)'s safe and effective messaging guidelines. The SPRC (https://sprc.org/), is a US government-funded non-profit organization that promotes collaboration among a wide variety of partners to build capacity in evidence-based suicide prevention strategies. Further, Jayman *et al.* (2023) concluded that 89% of young people in the UK felt pressure to be popular on social media, and 95% admitted that people were mean or unkind to each other on messaging on video-sharing platforms. Despite potential concerns with YouTube usage, its popularity could be leveraged to educate children and youth about online safety behaviors, and this can be achieved through public service announcements.

Public service announcements (PSAs) are effective instruments that raise awareness, educate society, and change individual, organization, community, and society behaviors and attitudes (Zhang *et al.*, 2017). Literature has documented the effectiveness of PSAs in communicating societal and health issues including texting while driving and healthy eating (Bummara and Choi, 2015; Zhang *et al.*, 2017). Moreover, PSAs published in social media now have larger audiences than those published in the traditional forms of radio, posters, and billboards, and they are more engaging due to their interactivity via shares, likes, and comments (Manganello *et al.*, 2020). Therefore, it is unsurprising that existing literature shows that PSAs have been used by many governments and organizations to promote online safety. For example, Amarah *et al.* (2020) reported about a group of local celebrities who teamed up with the Malaysian government to raise public awareness of cyberbullying and concluded that PSAs were an effective means to educate students. In addition, Waters *et al.* (2020) suggested that PSAs were appropriate for middle school teachers to educate their students on online kindness and avoiding cyberbullying. Finally, PSAs were viewed as one of the plausible approaches by the US government to warn individuals about cybersecurity attacks (Carlson *et al.*, 2022).

Despite the usefulness of PSAs in online safety targeting children and youth, there are two gaps in the current literature. The first gap is a lack of knowledge about the types of online safety topics that current PSAs address. There are individual research publications on online safety, such as the examples mentioned above, as well as PSA videos for raising awareness and fighting against skin color bias, hate, and harassment (DeVeaux *et al.*, 2020), PSAs for warning about online disinformation (Posard *et al.*, 2021), and those for building social cohesion among youth and eradicating racism (Madon *et al.*, 2022). What is missing is an inventory of the topics of online safety PSAs that have been created. Such an inventory of PSAs has benefits such as helping researchers, practitioners and policymakers identify and adapt suitable ones for their needs, perform comparative evaluations of effectiveness, and ascertain online safety topics that are relatively under-addressed.

The second gap is that there is limited understanding of how the content in PSAs is conveyed to people. Put succinctly, PSAs should be persuasive to their intended audience. Persuasion is an attempt to change attitudes using messages, focusing primarily on the characteristics of communicators and listeners (Boerman and Van Reijmersdal, 2020). The more persuasive appeals a PSA message applies, the more likely it will be accepted by its target audience (Pressgrove *et al.*, 2021). It should be noted that studies on the persuasiveness of individual PSAs on online safety have been conducted. These include the investigation of gain and loss frames (White *et al.*, 2018) and the effectiveness of PSAs in the form of narratives or arguments (Harris and Krishnan, 2023). However, like the first gap, a catalogue of the types of persuasive features employed by PSAs would be helpful in guiding research, practice, and policymaking in the context of online safety. Moreover, the current study is also interested in finding out how these persuasive features can engage the audience by eliciting different numbers of views, likes, and comments.

Given these research gaps, this study will address three research objectives. First, we ascertain the topics of online safety and their distribution that current YouTube PSAs cover. As mentioned earlier, YouTube is used in this study because it is one of the most successful video-sharing platforms with the second-highest number of active users worldwide after Facebook (Harper *et al.*, 2023; Zhaksylyk *et al.*, 2024) and is reported as the most popular social media platform among youth (Triliva *et al.*, 2015). Second, we determine the type of persuasive elements that these YouTube PSAs utilize for communicating with their audience. In our work, we will employ established persuasion theories to anchor our analysis. These theories will be discussed in the next section. Third, we determine the popularity of these PSAs by investigating the association between likes and comments, and the types of persuasive feature presented in them. These objectives will be achieved through a content analysis of YouTube PSAs. This method is appropriate because it permits researchers to describe the use of visual elements in YouTube videos in an objective, systematic, and quantitative manner for ease of understanding (Byrd-Bredbenner, 2002).

Literature review

Online safety

Online safety is a term that is commonly used yet infrequently defined. Used interchangeably with the terms "e-safety", "digital safety", and "internet safety", online safety usually refers to the protection of people who use devices and networks and is typically understood to include a range of overlapping risks relating to privacy, security, personal reputation management, and bullying and predation (Greyson *et al.*, 2023; Jang and Ko, 2023; Ma *et al.*, 2019).

Online safety can be understood from the other side of the same coin, that of online risks. For example, UK organizations that represent law enforcement, regulators, industry, and charities worked together to improve online safety in five prioritized areas – cyberbullying and sexual exploitation, radicalization and extremism, violence against women and girls, hate crimes and hate speech, and forms of discrimination against groups protected under the Equality Act (UK Council for Child Internet Safety, 2017). Further, Haynes and Robinson (2015) defined six main individual risks associated with social media sites: (1) external threats like phishing, identity theft, links to malware, and hijacking of personal sites, (2) stalking, harassment, and cyberbullying, (3) being targeted by criminals, for example, having one's home address published, (4) discrimination such as denial of health or life insurance or refusal of a job opportunity, (5) psychological harm like sharing personal information causing embarrassment with work colleagues and friends, and (6) persistent advertising such as spam.

Online safety is important to children and youth for three main reasons. First, as children and youth are learning to navigate the world, their media literacy is still under education, and they may lack the knowledge, abilities, and motivation to discern when they might be at risk of connecting to the Internet (Festl, 2021). Second, children and youth are developing their cognitive and social-affective capabilities more in the online environment due to intensive exposure to social media, which is accompanied by extensive changes in the structure and function of their adolescent brains (Crone and Konijn, 2018). This makes them more affected by the risks of the digital world due to their developmental vulnerabilities (Jagannath and Salen, 2022). Moreover, children and youth are often early adopters and innovators of new technology (Giovanelli *et al.*, 2020), but they may not be fully aware of the dangers around their use. This will likely create new vulnerabilities as well.

PSAs to promote online safety

Public service announcements (PSAs) are general-interest messages, which are primarily designed to inform and educate the public about a particular topic or issue (Jerome *et al.*, 2021).

PSAs take multiple forms, for example, posters, audios, and videos, and are disseminated to the target audience or specific segments of the population through various media channels, for example, print, radio, TV, or the Internet (Werb *et al.*, 2011). Their objectives are to bring about awareness and change public attitudes, opinions, or even behaviors toward an issue (Zhang *et al.*, 2017). These can be instructional, inspirational, or even shocking to elicit emotion and action (Searles, 2010). They have been applied to educate and engage children and young people on the potential risks associated with online activities, including cyberbullying, mental health, and inappropriate content (Amarah *et al.*, 2020; Samji *et al.*, 2022). By utilizing compelling narratives, relatable characters, and persuasive techniques, PSAs capture the attention of children and youth, encouraging them to adopt safe online practices (White *et al.*, 2018; Harris and Krishnan, 2023).

PSAs have been employed to promote online safety among children and youth. Hasinoff (2013) suggested that a PSA at the school morning assembly will be an easy measure for school administrators and teachers to handle students' concerns about sexting, harassment, and bullying. Yılmaz *et al.* (2017) also suggested that PSAs should be incorporated into school projects to improve safe Internet and computer use awareness of both teachers and students. PSAs were also used by government agencies from Malaysia and the USA to raise people's awareness of cyberbullying and cybersecurity attacks (Amarah *et al.*, 2020; Carlson *et al.*, 2022). These PSAs often employ emotions such as fear and efficacy, animations and stories with relatable characters, or statistics to attract children and young people's attention. Such attributes aim to encourage their comprehension and engagement. Collaborations between government agencies, educational institutions, and non-profit organizations have played a pivotal role in disseminating these PSAs through various platforms, especially via social media such as YouTube to cater to children and young people's habitual usage (Bator and Cialdini, 2000). For example, the Singapore Police Force, Tote Board Singapore, and National Crime Prevention Council collaboratively produced an online love scam story, where a young lady named Ella was scammed out of S$25,000 by Brandon, a pig butchering scammer (National Crime Prevention Council, 2020).

Message framing

Since PSAs have the potential to influence people's attitudes and intentions on public issues (Lee *et al.*, 2018), their messages are worthy of examination for successful public safety campaigns and communication effectiveness. Message framing refers both to the process of selecting and the way information is presented (Wicks, 2005; Nab *et al.*, 2020). Message framing is an integral part of conveying and processing data daily, and successful message framing can reduce the ambiguity of intangible topics by contextualizing the information in such a way that recipients can connect to what they already know (Swift and Peterson, 2019).

Levin *et al.* (1998) proposed a typology distinguishing three types of framing: attribute framing, goal framing, and risky-choice framing. Risky-choice framing is the most studied message frame, widely applied in psychology, health, and medicine (Xu and Huang, 2022; Hsu, 2023). It presents outcomes in terms of gains or losses (Tversky and Kahneman, 1981). This may lead to changes in risk-taking willingness as people tend to take risks to avoid losses rather than to achieve gains. This tendency is explained by prospect theory's psychophysical value function, which is concave for gains and convex for losses (Kahneman and Tversky, 2013).

Prospect theory

Prospect theory has received much scholarly attention in message-framing research (Van't Riet *et al.*, 2016; Huang and Liu, 2022). It explains how message framing influences an individual's decision-making. The theory suggests that (1) people may prefer less risky outcomes when exposed to messages involving potential gains (positive consequences) and (2) people tend to seek risky outcomes when exposed to messages involving potential losses (negative consequences) (Huang and Liu, 2022). In prospect theory, risk is conceptualized as uncertainty and deals with probability: high risk equates to more uncertainty, and low risk equates to more certainty (Harrington and Kerr, 2017; Chen *et al.*, 2022). When facing gain-framed messages, people tended to seek low-risk (certain) outcomes, and when facing loss-framed messages, people preferred high-risk (uncertain) outcomes, providing empirical support for prospect theory (Tversky and Kahneman, 1981).

Prospect theory is also widely used in PSA research. Sivakumaran *et al.* (2023) conducted a literature review on PSAs and found that prospect theory was one of the most frequently theories used. Moreover, they found that besides simply stating issues, it is critical for PSAs to demonstrate the steps of how to address them (Sivakumaran *et al.*, 2023). Kim and Baek (2023) found that loss-framed messages were more effective in generating unfavorable attitudes toward texting while driving than gain-framed messages among participants. Lastly, Mayer *et al.* (2022) found that positive language and positive frames on childhood obesity elicited more positive emotional, attitudinal, and behavioral intent outcomes.

Exemplification theory

Exemplification theory suggests that in each message, the use of exemplar characters embedded in a narrative will be perceived as more engaging and persuasive than the use of factual evidence such as statistics (Zillmann, 2006). Narrative evidence uses exemplars - stories about an individual's experiences to grab the attention of the audience, engage them, and influence their opinions via bringing "a human face or an emotional angle to the presentation of an event issue, or problem" (Semetko and Valkenburg, 2000, p. 95). Audiences value exemplars because they serve as a point of comparison with their own lives (Petraglia, 2009; Young *et al.*, 2021). In contrast, statistical evidence presents information in a factual manner, with percentages and numbers to support a conclusion or offer a summary record across large populations (Allen *et al.*, 2000). Some need-for-cognition-oriented individuals would prefer logically presented facts during their information processing and decision-making (Xiao *et al.*, 2023).

The effectiveness of narrative versus statistical evidence is widely used in PSA research. Reinhart and Lilly (2022) found that narrative PSAs on organ donation elicited more positive message reactions than statistical PSAs. Conversely, a PSA with a narrative focus did not elicit a greater intention to communicate with loved ones about donations than a PSA with a statistical focus. However, Zhang *et al.* (2017) found that statistical evidence was more likely to appear in PSAs promoting healthy eating behavior, especially for the costs and the dangers of not eating healthily when compared to narrative evidence. Further, Limon and Kazoleas (2004) explored the use of exemplars versus statistical evidence to reduce the number of counterarguments and overall responses to a PSA on the dangers of tanning and the use of tanning beds. They found that narrative evidence produced fewer counterarguments and fewer responses than statistical evidence although they found no difference in the persuasiveness of the two evidence types.

Methodology

Sample

Data collection was conducted between 1 to 23 March 2023. The combinations of phrases "online safety" or "e-safety" "digital safety" "public service announcement" "public service message", "PSA", and "children and youth", or "teens" and "adolescents" or "youth" "young people" were entered into YouTube's search engine to retrieve videos. The investigators went through the first 12 results pages, each with 20 videos. As the subsequent pages included more duplicates and the topics of the videos became more unrelated, data collection was confined to the first 12 pages. To be included in the sample, videos had to meet the following requirements: (1) longer than 30 s, as videos that are too short will be less informative for a PSA, (2) in English, (3) suitable for children and youth, and (4) related to online safety and its subcategories, for example, cyberbullying, mental health, cybersecurity or online safety behavior. Videos were screened out if they were (1) not made for children and youth, (2) more than 5 min, and (3) not in the English language. Further, the videos were checked for duplicates regardless of the title. After removing three videos less than 30 s, four videos longer than five minutes, and 13 duplicates, a total of 220 videos remained. For each of these videos, the posting date, length, number of views, number of likes, and number of comments, together with the title and the URL were recorded.

Procedure

Content analysis is "a research technique for the objective, systematic and quantitative description of the manifest content of communication" (Drisko and Maschi, 2016, p3). Historically, it has been used to analyze and draw inferences from images and text, but it has been extended to other forms of media, including video (Morgan *et al.*, 2010). Krippendorff (2004) explained that "… works of art, images, maps, sounds, signs, symbols, and even numerical records may be included as data - that is, they may be considered as texts …" (p. 19).

The unit of analysis was each video. Two coders pretested and refined the codebook. First, the sponsor of the video was coded as either affiliated to one of three types of organizations: government organizations, not-for-profit organizations, for-profit organizations, or individuals who are not affiliated to any business entity. Sponsor information was obtained using either the credits at the end of each video, the logo, or from the YouTube information section. If no source information could be identified, the video was deemed to be created by an individual owner of the account, and hence coded as an individual sponsor. This approach was adopted by Zhang *et al.* (2017) who coded PSA video sponsors similarly.

Next, the themes of the videos were coded. Combining the taxonomy of online risk (Haynes and Robinson, 2015) and the Internet Code of Practice (IMDA, 2016), with the pretested results of the codebook, six major themes were generated: (1) general instructions on online hygiene, (2) online safety behavior/conduct, (3) online public security instructions, (4) mental health/wellness, (5) online hate and harassment, (6) sexual abuse and exploitation (see Table 1). If one video had two or more themes at the same time, the main theme was ascertained and coded as such.

Thirdly, the videos were coded according to the theoretical frameworks used. Using prospect theory, message frames were coded as gain, loss, or neutral. When a video emphasized methods to enhance or protect one's online safety such as through following guidelines, it was considered as gain framed. When the video mostly described a negative consequence of an action or the actors in the video were put in danger, the video was considered as loss framed. When there was neither an emphasis on safety enhancement nor the risk of danger, the video was neutrally framed, such as only recommending steps for audiences to follow without mentioning positive or negative consequences. Next, using exemplification theory, videos were coded as messages with narrative evidence when an individual story or collection of stories of online experiences was presented either from a first-person or a third-person perspective. Messages were coded with statistical evidence if the information was presented factually with percentages and numbers, while messages with narratives and statistics were coded with both. Finally, messages were coded with no evidence if no stories or statistics were presented.

Themes	Explanation with examples
General instructions on online or cyber hygiene	Messages focusing on maintaining the health and security of devices, networks and data. Examples include strong passwords, updating software, protection from malware and hackers
Online safety behavior/conduct	Messages about maintaining the social norms of interactions in private and public online spaces. These include being nice, respecting disagreement, keeping away from online strangers, and limiting time spent online
Online public security instructions	Instructions on online public security, especially on phishing and scams. For example, ID theft and misrepresentation, and bogus documents such as shopping invoices and deliveries
Mental health/wellness	Content dealing with mental health/wellness, encompassing psychological, emotional, and social health. Examples are stress management and fostering good relationships with other people
Online hate and harassment	Messages warning against online hate and harassment, encompassing negative comments that likely to cause harassment, alarm or distress to the target person or group by using threatening, abusive or insulting words or behavior. Examples are online bullying comments or behaviors, doxing and stalking
Sexual abuse and exploitation	Messages warning against sexual abuse and exploitation. Examples can be unsafe behaviors, explicit or implicit contents that may result in sexual abuse, sexual exploitation, grooming or any online activity for sexual gratification

Source(s): Table by authors

Table 1. Themes with explanation

One author trained an undergraduate student on the coding protocol which included the coding scheme, operational definitions of each coding category, and coding procedures. The author and the student then independently viewed and coded the same 20% random sample of videos (44 of 220). The uncertainties and disagreements during this stage were marked, discussed, and resolved.

These included the themes, types of frames, and evidence. Subsequently, the rest of the 176 videos were equally divided among the two coders for independent coding. Cohen's kappa was used to measure the intercoder reliability of the 176 videos. A kappa value between 0.61 and 0.80 is considered substantial, and more than 0.80 is considered an almost perfect agreement (Savanović *et al.*, 2023). The kappa results were all above the minimum requirements and are as follows: sponsoring organization ($K = 0.797$), themes ($K = 0.786$), gain frame ($K = 0.765$), loss frame ($K = 0.767$), neutral frame ($K = 0.799$), narrative evidence ($K = 0.781$), statistical evidence ($K = 0.805$), both evidence ($K = 0.821$), and no evidence at all ($K = 0.901$).

Results

Video characteristics

To address the first research objective, the characteristics of videos in terms of length, publication time, number of views, number of likes, number of comments, sponsor types, country of origin, and themes were analyzed. More than 60% of the videos were between one to three minutes long, while 25% were three to five minutes in length. The remainder were under a minute. In terms of age, 48% were published within the past four years, 41% between four to 10 years, and the remainder more than 10 years.

Viewership numbers of the videos varied widely. The majority (57%) had under 1,000 view counts, 33% had between 1,000 to 99,999 view counts and 10% of the videos were viewed at least 100,000 times, with the maximum being 906,505. For likes attracted, more than half of the videos attracted less than nine likes, 23% of them attracted between ten to 99 likes, and 20% attracted more than 100 likes, with the maximum being 381. For comments, one-quarter of the videos turned off this function. Most (38%) had no comments, 29% of the videos attracted one to 99 comments, 6% attracted 100 to 499 comments, and 2% attracted 500 to 999 comments, with the maximum being 636.

Table 2 shows the distribution of sponsors. Businesses published the most PSAs followed by government organizations, individuals, and non-profit organizations. Regarding the origin countries (Table 3), the US published the most online safety PSAs in our sample, with the UK, Australia, and Canada trailing far behind. The origin of around 22.27% of the videos could not be identified. Other countries that published one to two videos were bundled into one category: two videos were published in New Zeeland, two in South Korea, two in South Africa, one in Zimbabwe, one in Finland, one in Czechoslovakia, one in Hungary, one in Brazil, one in Austria, one in Sweden and one in France.

Table 4 shows the distribution of themes in our sample of videos. PSAs about online safety behavior were the largest category at 30.91%, followed closely by online hygiene PSAs. Videos about sexual abuse and exploitation occurred least frequently.

Distribution of gain-loss frames

Figure 1 presents the distribution of videos coded by frame type. Overall, more than half of the videos (63.18%; 139 of 220) adopted a neutral stance. For example, animated characters Jorge and his sister Hanna are active online. They showed people steps to protect personal information, for example, making sure there is a lock icon when Jorge buys music or concert tickets online, and always logging out after using public computers. There was no mention of gains or losses due to the actions described. Next, gain-framed (18.64%; 41 of 220) and loss-framed videos (18.18%; 40 of 220) were almost equally distributed. An example of the former

Table 2.
Types of sponsors

Sponsors	Percentage (# of videos)
Businesses	46.82 (103)
Governments	25.91 (57)
Individuals	18.18 (40)
Non-profit organizations	9.09 (20)

Source(s): Table by authors

Table 3.
Country distribution

Country of publisher	Percentage (# of videos)
USA	48.64 (107)
Unknown	22.27 (49)
UK	5.46 (12)
Australia	5.46 (12)
Canada	5.00 (11)
UAE	2.27 (5)
Singapore	2.27 (5)
India	2.27 (5)
Other countries	6.36 (14)

Source(s): Table by authors

Themes	Percentage (# of videos)	Example
Online safety behavior/conduct	30.91 (68)	An animated character describes three usual practices online, (1) be nice and respect disagreement, (2) only befriend people you know in real life, (3) do not meet strangers
General instructions on online hygiene	28.18 (62)	A teenage boy shares about how to protect oneself online, including using strong passwords, using https rather than http, updating software and using VPN to safeguard the connection
Online hate and harassment	16.36 (36)	A young man named Terrence kept posting negative comments about his friends, which attracted other people to join him. Meanwhile, Terrence's sister Emily was called a loser online by her friends and was told to die. Emily went through extreme emotions and depression. Terrence noticed Emily's emotions and checked her phone. He realized the negative consequences in real life of his online posting and apologized online to the victim
Mental health and well-being	12.73 (28)	A young man was addicted to playing online games day and night until a teacher called his parents
Instructions on online public security	10.00 (22)	A university student shows how to avoid online scams such as phishing, including checking for legitimate email address if receiving a suspicious message and checking credit card bills regularly in case of any suspicious payments
Sexual abuse and exploitation	1.82 (4)	A young woman was seeking a male friend' comfort online. The man asked for her nude picture to prove her true love for him, and promise won't abuse it. The woman accidentally sent her picture to the public, even though she deleted the picture immediately, someone still snapshot it. She became the focus of attention and gossip

Source(s): Table by authors

Table 4. Distribution of themes

involved three young people who were interviewed about how to stay safe online and avoid scams by using strong passwords for different accounts and not clicking any links for quick money. Next, an example of a loss-framed video involved several children and young people offering their perceptions of oversharing and how doing so could result in stalking or kidnapping.

Adopting a frame-oriented perspective, we examined the distribution of themes among each frame type, which is depicted in Figure 1. For neutral-framed videos, online safety behavior and online hygiene were the two most frequently presented themes, each taking 31.65% (44 of 139), followed by mental health and well-being (15.83%; 22 of 139), online hate and harassment (14.39%; 20 of 139). The least mentioned themes within neutral-framed videos were online public security (6.47%; 9 of 139), and sexual abuse and exploitation (0%; 0 of 139). Likewise, among the gain-framed videos, online safety behavior and online hygiene were the two most frequently presented with 31.71% for the former (13 of 41) and 29.27% for the latter (12 of 41), followed by 21.95% on online public security (9 of 41), 9.76% on mental health and well-being (4 of 41), and 7.32% on online hate and harassment (3 of 41). There were no gain-framed videos on sexual abuse and exploitation. In contrast, loss-framed videos had a different pattern of theme distribution. Here, 32.50% of them (13 of 40) presented the theme of online hate and harassment, and 27.50% (11 of 40) on online safety behavior. Less frequently mentioned themes were online hygiene (15.00%; 6 of 40), online public security, and online sexual abuse and exploitation (10.00%; 4 of 40 for each). Mental health and well-

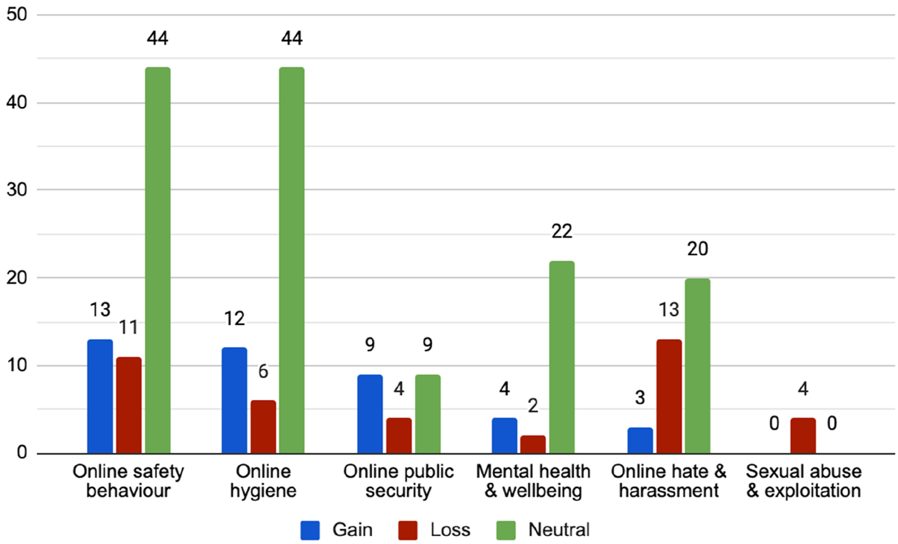


Figure 1.
Distribution of frames over themes

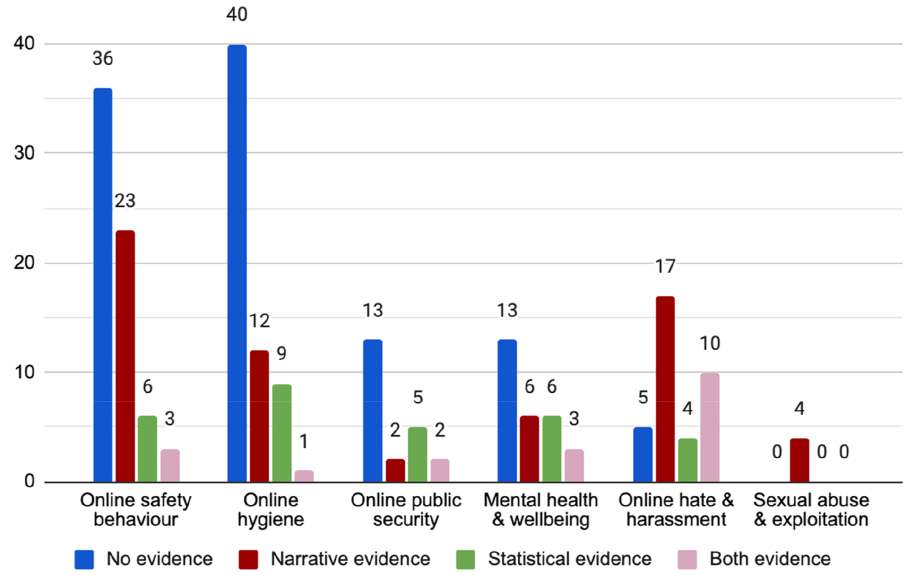
Source(s): Figure by authors

being (5.00%; 2 of 40) was least mentioned. Adopting a theme-wise perspective, videos on online safety behavior, online hygiene, and mental health and well-being shared the same pattern in frame distribution, that is, more neutral-framed videos than gain-framed ones, and even more than loss-framed videos. While gain and neutral frames were equally distributed among online public security videos (40.91%; 9 of 22), the rest were loss-framed videos (18.18%; 4 of 22). Surprisingly, videos on online hate and harassment and sexual abuse and exploitation had more loss-framed videos than other types. This comprised 36.11% for the former theme (13 of 36) and 100% for the latter (4 of 4). This was unexpected because, in prior research, gain-framed messages in anti-bullying campaigns were perceived as more favorable and effective than loss-framed ones (Bright, 2015).

Distribution of exemplification appeals

Figure 2 shows the distribution of exemplification appeals. Almost half of the videos (48.64%; 107 of 220) presented no evidence. One typical example is a female voice describing three steps to stay safe online, including never sharing your personal information, using strong passwords, and only shopping on known websites. Next, 29.09% of the videos (64 of 220) presented narrative evidence. One example shows four college students being interviewed about their experiences with cyberbullying and its negative consequences. Further, 13.64% of the videos (30 of 220) offered statistical evidence, such as another cyberbullying video stating that as many as one in three Australian children experienced it in some form. Finally, a few videos (8.64%, 19 out of 220) employed both narrative and statistical evidence. One such video mentions that there were 2.77 billion chances for cyber-attacks because there were 2.77 billion social media users worldwide in 2019. Following this, a young man tells his story about how he set various privacy settings on his mobile to protect his personal information.

Figure 2 shows the distribution of exemplification appeals. For those videos providing narrative evidence, online safety behavior (35.94%; 23 of 64) and online hate and harassment (26.56%; 17 of 64) were the most frequently occurring, followed by online hygiene (18.75%;



Source(s): Figure by authors

Figure 2. Exemplification appeals distribution over different themes

12 of 64), mental health and well-being (9.38%; 6 of 64), sexual abuse and exploitation (6.25%; 4 of 64), and online public security (3.12%; 2 of 64). Next, for videos giving statistical evidence, 30.00% were on online hygiene (9 of 30), and 20.00% were on online safety behavior and mental health and well-being respectively (6 of 30). This was followed by online public security (16.67%; 5 of 30), and online hate and harassment (13.33%; 4 of 30).

For videos presenting both types of evidence, 52.63% of them focused on online hate and harassment (10 of 19), followed by online safety behavior and mental health and well-being (15.79%; 3 of 19 for each), and online public security (10.53%; 2 of 19). Online hygiene was the least frequent (5.26%, 1 of 19). Finally, for videos giving no evidence, 37.38% of them (40 of 107) were on online hygiene, 33.64% were about online safety behavior (36 of 107), followed by online public security and mental health and well-being (12.15%; 13 of 107 for each). Online hate and harassment (4.67%; 5 of 107) occurred least frequently.

Theme-wise, more than half of the videos on online hygiene (65.57%; 40 of 62), online public security (59.09%; 13 of 22), and online safety behavior (52.94%; 36 of 68) provided no evidence at all. Next, about half of the videos on online hate and harassment (47.22%; 17 of 36) and all sexual abuse and exploitation (100%; 4 of 4) provided narrative evidence. About one-fifth of the videos on online public security (22.73%; 5 of 22) and mental health and well-being (21.43%; 6 of 28) provided statistical evidence. Around one-third of the videos on online hate and harassment provided both types of evidence (27.78%; 10 of 36), followed by online public security (13.64%; 3 of 22), mental health and well-being (10.71%; 3 of 28), online safety behavior (4.41%; 3 of 68), and online hygiene (1.61%; 1 of 62). Finally, no sexual abuse and exploitation videos provided both types of evidence.

Engagement with audiences

Correlational analyses were conducted between the video length and the number of views, likes and comments gathered. We found that video length was weakly associated with the number of views it gathered ($r = 0.171$, $p < 0.05$) and the number of likes ($r = 0.164$, $p < 0.05$), but not with the number of comments ($r = 0.122$, $p = 0.106$). Specifically, longer videos attracted more views and likes than shorter ones, but there was no difference between them in attracting comments.

Next, Chi-square tests were conducted for sponsorship, themes, frame appeals, and use of evidence appeals. We found that the sponsorship of the videos did not significantly influence the number of comments (χ^2 (3) = 0.226, $p = 0.878$) or the number of likes (χ^2 (3) = 0.220, $p = 0.883$), although there was a significant influence on the number of views gathered (χ^2 (3) = 3.958, $p < 0.01$). Specifically, videos published by government organizations received more views than those published by businesses, non-profit organizations, and individuals. Interestingly, other video characteristics did not significantly influence audience engagement. These were video themes, frame appeals, and evidence appeals. Put differently, the number of views, likes and comments did not vary according to the video theme or how a video was framed.

Discussion

The purpose of this study was to uncover the topics related to online safety in YouTube videos, ascertain the persuasive elements utilized and determine the relationship between their characteristics and persuasive features and audience engagement metrics. YouTube videos provide producers with the opportunity to discuss online safety and engage millions of people around the world. Given our findings, there are four points worth noting.

First, our findings revealed that businesses were the leading sponsors of online safety PSA videos. This could be because of the nature of the YouTube platform. YouTube was initially developed as a digital information and entertainment channel but later became a significant revenue-generating channel through marketing communication (Febriyantoro, 2020), and it is an effective tool to increase one's brand image and strengthen brand awareness. Business-sponsored PSA videos usually provide step-by-step instructions to safeguard personal information and account safety, aiming for the good of the public. At the end of such videos, the company name and contact information are shown for brand awareness and encouragement for public inquiries. Moreover, it appears that the majority of viewers of these videos affirm the credibility of the content and advocate for other people to learn from them. For example, one of the comments under an online hygiene PSA recommending using two-factor authentication said that "Every beginner should watch this video. And yeah, I also say that you need to use two-factor authentication on every platform you have." By raising public awareness and educating them to follow the instructions on online safety, these PSAs published by businesses provide an important service to viewers as well as raise their profile on YouTube.

Next, our findings showed that online safety behavior and online hygiene were the main themes among the online safety videos, while public security and sexual abuse and exploitation were the least frequently produced. The importance of the former two may be explained by their foundational roles in the network society. In online social interaction, online safety behavior is the first and foremost thing that people should learn when they engage with others so that they can adequately protect themselves (Park *et al.*, 2014). This is similar to the best practice of all drivers who should pay attention to road safety when they start the engine (Jane, 2017). Online hygiene - maintaining the security of devices, networks, and information, is similarly important and should be properly integrated into simple routines to raise individuals' online security awareness (Ma *et al.*, 2019). Given that these fundamental topics were relatively easy to understand across age groups (F5 Lab, 2020), such as avoiding strangers online, and were not controversial, it was perhaps also unsurprising that there were more such videos produced.

Interestingly, with the rise of online scams and increasing concerns about cybersecurity and sexual abuse, these two categories of videos were the two smallest categories of videos encountered in our sample. This is probably due to two reasons. First, compared to video-based methods in promoting security awareness among the public, several other approaches could be more effective. Prior research indicates that games and tutorials, designed to enhance anti-phishing skills, are particularly effective for children and young adolescents compared to PSAs (Baslyman and Chiasson, 2016; Williams *et al.*, 2019). For instance, in an experiment by Sheng *et al.* (2007), participants who played an interactive anti-phishing game were more adept at identifying fraudulent websites than those who read an anti-phishing tutorial or existing online training materials. Additionally, given the widespread and critical issue of sexual abuse and exploitation of children and young people (Vosz *et al.*, 2023), school-based educational programs have become the most common prevention strategy (Bright *et al.*, 2022). These programs often encompass a variety of activities, such as group learning, skill-building activities, role-play, and collaborative activities (Brecklin and Forde, 2001). The diverse and engaging nature of these activities tends to dilute the importance and quantity of online safety videos specifically designed for educating children and youth about sexual abuse and exploitation.

Second, creating videos to address the concepts of sexual abuse presents several challenges. Research highlights that discussing polarizing, controversial, and sensitive topics can be stressful for youth, regardless of their prior experiences with victimization (Sætra, 2021). Given that such videos may contain sensitive imagery and text, there is a risk that inappropriate content could increase students' anxiety or fear. Consequently, it is recommended that videos should be carefully crafted in content, positively framed, and avoid using scare tactics. Additionally, the delivery length and frequency of these videos should be thoughtfully considered to maximize their educational effectiveness (Onrust *et al.*, 2016; Lemaigre *et al.*, 2017).

Finally, the majority of the PSA videos adopted a neutral frame, while the gain-framed and loss-framed videos were equally distributed in comparatively smaller proportions. Meanwhile, about half of the PSA videos provided no evidence in support of the content at all. One possibility for the absence of persuasive appeals could be due to the lack of knowledge of message framing principles and relevant theories by video creators, such as businesses, individuals, and non-profit organizations (Thomke, 2023). However, research has demonstrated that it is essential to include theoretical guidance, such as knowledge related to persuasion, in strategic communications to improve public awareness of issues (Bator and Cialdini, 2000). As many of our PSA videos lack theoretical guidance, there is a risk that they are rendered less effective in their communication with children and youth.

Conclusion

The following implications may be drawn from our study. First, this work contributes to prospect theory and exemplification theory. Since more than half of our PSA videos adopted a neutral frame and without explicit evidence, our research extends the application of these theories in the creation of PSA videos. By analyzing the content of online safety PSA videos, our work provides an important example of how these theories could be applied to children's and youth's online safety education.

Second, this study enriches the online safety literature by conducting a content analysis of PSA videos, specifically targeting children and youth. Our findings, which include the characteristics of these videos, their persuasive elements, and the relationship between video characteristics and audience engagement, provide valuable insights into online safety education. These results can serve as a guide or starting point for future research on online safety education for children and youth, as well as on the persuasiveness of PSA videos.

Our results also offer the following practical implications. First, we recommend that producers create videos lasting between one and three minutes. Our findings suggest that video length plays a significant role in audience engagement, as indicated by the number of views and likes. This duration is long enough to convey information yet short enough to capture attention and encourage engagement. Furthermore, it is highly advisable for producers to incorporate theoretical elements such as those drawn from prospect and exemplification theories into their content production. It is crucial to recognize that the theoretical foundations of these appeals can enhance the educational value and relevance of the content. Employing these theories can also assist in structuring messages to maximize educational outcomes, thereby potentially increasing public awareness.

Second, we advocate for countries beyond the USA to publish more online safety PSAs that are tailored to their specific cultural contexts. This is essential to raise citizens' awareness and educate them on online safety issues. Online safety is a global and pervasive issue, as two-thirds of the global population is connected to the Internet, a figure that is expected to grow (Petrosyan, 2023; Sund, 2007). However, our findings reveal that almost half of the videos we analyzed are produced in the USA, indicating a significant gap that other countries could fill. Considering the critical importance of this topic, governments, policymakers, and related organizations in other countries should give greater attention to online safety issues and increase their production of informative videos.

Limitations and future work

This study is not without limitations. First, we only sampled online safety videos from YouTube and our findings may not be generalizable outside this platform. There are other popular video-sharing social media platforms, such as Instagram and TikTok, among children and youth. Online safety videos on those platforms may exhibit different patterns of PSA characteristics. Second, our work was informed by prospect and exemplification theories. Although we found a large number of videos that were seemingly atheoretical, it could be the case that they were guided by other theoretical frameworks not covered in this research. Third, our work examined the number of views, likes, and comments as metrics for audience engagement. While these metrics have been used in related prior research (Tenenboim, 2022; Hiaeshutter-Rice and Weeks, 2021; Munaro *et al.*, 2021), we note that relying solely on numbers may not accurately reflect the quality or type of audience engagement. A missing element is the analysis of the comments themselves, which may provide information on how audiences perceive and react to these videos.

For future studies, and in line with the first limitation, it is recommended that online safety PSA videos can be sampled from platforms other than YouTube such as Instagram and TikTok. Comparative studies can be conducted to investigate the different patterns of themes, frames, and the extent to which these videos engage with their audiences. Further, it would be worthwhile to investigate cultural differences and how they impact the content presented in such videos. Next, persuasive techniques drawn from other models could be studied, for example, the Extended Parallel Process Model, which is one of the most widely applied models for persuasion research (Shirahmadi *et al.*, 2020). Moreover, as mentioned, this research focused only on the video content. Future complementary work could analyze the comments associated with each video to uncover audiences' attitudes toward the videos.

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About the authors

Liuyu Huang is Research Associate at the Wee Kim Wee School of Communication and Information, Nanyang Technological University, Singapore. Her research interests include bibliometrics, information studies. Liuyu Huang is the corresponding author and can be contacted at: lyhuang@ntu.edu.sg

Dion Hoe-Lian Goh is Professor at the Wee Kim Wee School of Communication and Information, Nanyang Technological University, Singapore. His research interests include social media practices and perceptions, gamification techniques for shaping user perceptions and motivating behavior, mobile information sharing and seeking, and serious games for health and education.

Stella Xin Yin is Ph.D. candidate at the Wee Kim Wee School of Communication and Information, Nanyang Technological University, Singapore. Her research interests include computational thinking, collaborative learning, and AI literacy.